

Dynamic, Task-Aware Construction of Knowledge Graphs from Text

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1. Research Problem

This project investigates the construction of task-aware knowledge graphs (KGs) from text by treating graph structure as an optimization target rather than a fixed design choice. Existing approaches typically generate large, general-purpose graphs in which ontology and structure are specified a priori and evaluated independently of their downstream use. While such methods maximize coverage, they often produce representations that are noisy, redundant, and poorly aligned with predictive objectives.

In contrast, this project introduces a feedback-driven framework in which knowledge graph construction is explicitly conditioned on task performance. Ontology is treated as a tunable component, and graph structure is iteratively refined based on signals derived from downstream models. Using document-based datasets (e.g., financial news linked to stock movements or biomedical literature linked to treatment outcomes), large language model (LLM) agents generate and evolve ontologies and extract structured representations, producing multiple candidate graph variants.

The central problem is therefore not only how to extract knowledge from text, but how to construct graph representations whose structure is optimized for a specific task while remaining adaptive to continuously evolving data.

2. Research Questions

1. To what extent does task-aware knowledge graph construction improve predictive accuracy in node-level classification tasks (e.g., price movement prediction) compared to static graph construction approaches?
2. Can task-aware optimization produce smaller, more efficient knowledge graphs while maintaining or improving downstream task performance?
3. Does iterative, task-driven knowledge graph construction lead to more informative graph structures and enable the identification of effective ontology configurations?

3. Theoretical Framework

The project is grounded in knowledge representation and representation learning, but departs from standard formulations by conceptualizing knowledge graphs as task-dependent and dynamically optimized structures. Central to this perspective is a

reinterpretation of ontology: rather than a fixed schema, it is treated as a hyperparameter that governs both the structure of the graph.

This view leads to a closed-loop, feedback-driven paradigm. Multiple knowledge graph variants are generated under different ontology configurations and evaluated through downstream predictive tasks. The resulting performance signals are then used to iteratively refine the ontology and construction process, aligning the representation with task-specific objectives.

In the financial application, this framework is extended to incorporate temporal dynamics. Rather than relying on globally stable embeddings, time is represented through local graph structure, where each time step corresponds to a distinct subgraph capturing the current informational neighborhood. This allows the model to capture changes in relationships and context over time while maintaining comparability across observations.

4. Data, Sources, and Methods

The empirical analysis relies on document-based datasets that link textual information to measurable outcomes. The primary application focuses on financial news corpora—comprising articles, reports, and sentiment data—paired with stock market movements to perform node-level price movement classification. Specifically, the project utilizes datasets such as the FNSPID (Financial News Sentiment and Price Intensity Dataset), focusing on entities like NASDAQ-listed companies. While focused on finance, the framework is extensible to other domains, such as linking biomedical literature to adverse drug reactions.

Knowledge graphs are constructed using a multi-agent system: an Ontology Agent proposes and updates the schema, while a KG Builder Agent performs structured extraction of entities and relations. This process identifies a wide range of domain-specific entities—such as Companies, Analysts, Deals, Products, and Markets—and captures their interactions through relationships like "ASSOCIATED_WITH", "INVOLVES", and "TRACKS".

The methodology proceeds by treating the ontology as a hyperparameter, generating multiple candidate KGs under varying configurations. For each instance, subgraph embeddings and other graph-derived features are computed and integrated into predictive models, such as logistic regression and gradient boosting. A closed feedback

loop is established: downstream model performance (measured by metrics like AUC and F1-score) provides direct feedback to the Ontology Agent, enabling iterative refinement of the graph construction process.

5. Risks

The project involves several conceptual and implementation-related risks. Conceptually, optimizing graph structure for a specific task may lead to over-

specialization, reducing generalizability across domains or tasks. Additionally, task-driven selection may omit information that is important but only weakly predictive.

From an implementation perspective, the approach depends on the reliability of LLM-based extraction, which is non-deterministic. The need to generate and evaluate multiple graph variants also creates computational challenges, particularly in iterative settings.

6. Ethical Considerations

The proposed framework raises concerns related to bias, transparency, and domain-specific impact. Biases in source texts or introduced during LLM-based extraction may be encoded and amplified in the constructed graphs, particularly when optimization is driven by predictive performance.

If deployed, such systems would inherit and propagate these biases. In high-stakes domains such as finance or healthcare, this may lead to systematically distorted or harmful outcomes.

Possible mitigation strategies include bias auditing, careful data selection, and transparent, interpretable evaluation.

7. Budget Categories

- (Possibly) Personnel (research and engineering support)
- Cloud computing and infrastructure (including graph databases and distributed processing)
- Data acquisition and licensing (e.g., financial and biomedical datasets)
- Software and development tools
- Dissemination (conferences, publications, workshops)